

Wenlai Han

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RESEARCH INTERESTS	Skeletal and peripheral remodeling of C(sp ³)-rich compounds under dynamic kinetic control; late-stage functionalization and the differentiation of parallel reaction pathways through reaction-network control; selective constitutional isomerization and fluxional behavior; metal–ligand cooperativity in catalysis; and the development of automated reaction-discovery pipelines.	
EDUCATION	Massachusetts Institute of Technology , Cambridge, MA (expected 2028) Ph.D. Candidate, Organic Chemistry · <i>GPA: 5.00/5.00</i> XtalPi AI for Science Fellow; Advisor: Prof. Alison Wendlandt Carleton College , Northfield, MN B.A. in Chemistry and Computer Science; minor in Music Performance. <i>Magna cum laude</i> with distinction in the chemistry major · <i>GPA: 3.92/4.00</i> Advisors: Prof. Matt Whited, Prof. David Alberg, Prof. Jeff Ondich Stanford University , Stanford, CA High School Summer Visiting Student · <i>GPA: 3.93/4.00</i> China Agricultural University , Beijing, China Visiting High School Research Assistant (during academic breaks)	2023 – Present 2019 – 2023 2018 2016 – 2018
JOURNAL PUBLICATIONS	“Synthesis and Cooperative Reactivity of Sterically Encumbered Silyl and Silylene (^t BuP ₂ Si)Co Complexes”, Whited, M*; Pearson, C; Stewart, M; Lee-Zacheis, C; Khan, A; Han, W ; Giuliani, K; Kohen, D, <i>Organometallics</i> 2026, 45, 3, 297–305. [doi] “Migrating Group Strategy for Remote Functionalization of Seven-Membered Rings”, Han, W [†] ; Hwang, T [†] ; Lian, C; Kolb, S; Yamane, M; Palani, V; Wendlandt, A*, <i>J. Am. Chem. Soc.</i> 2025, 147, 35, 32077–32084. [doi] “Cobalt Silylenes as Platforms for Catalytic Nitrene-Group Transfer by Metal-Ligand Cooperation”, Whited, M*; Han, W ; Jin-Lee, H; DiNardo, Z; Watson, E; Zhang, J; Kohen, D, <i>Angew. Chem. Int. Ed.</i> 2022, 61, e202205748. (Hot Paper) [doi] “Novel Application of CO ₂ -Assisted High Pressure Processing in Cucumber Juice and Apple Juice”, Zhao, L; Qin, X; Han, W ; Wu, X; Wang, Y; Hu, X; Ling, J; Liao, X*, <i>LWT</i> 2018, 96, 491–498. [doi]	
SELECTED CONFERENCES & PRESENTATIONS	“Migrating Group Strategy for Remote Functionalization of Seven-Membered Rings”, Han, W ; Hwang, T; Lian, C; Kolb, S; Yamane, M; Palani, V; Wendlandt, A, <i>Amgen Young Investigator Award Symposium</i> and <i>MIT–Merck Symposium</i> , 2025. “Cobalt Silylenes as Platforms for Catalytic Nitrene-Group Transfer by Metal-Ligand Cooperation”, Han, W ; Whited, M, <i>Gulf Coast Undergraduate Research Symposium</i> . Outstanding Presentation in Inorganic Chemistry Award and Travel Award, 2022. “Metal-Ligand Cooperative Transfer of Alkyl Azide to Isocyanate Facilitated by a Si=Co Platform”, Han, W ; Whited, M, 263 rd <i>ACS National Meeting</i> . ACS–DIC Symposium Poster Prize and ACS–DIC Travel Award, 2022.	

COMPUTATIONAL & SOFTWARE PROJECTS	• PocketForge – a full-stack design–make–test–analyze (DMTA) drug-design pipeline with validated data APIs and explainable selectivity scoring. • Know-de – an agentic tutor that guides students through Socratic questioning rather than direct explanation. • Geothermal Data Web App – an interactive data dashboard built for Carleton Geology. • PM_{2.5} Sensor Pipeline – data processing and analysis for low-cost passive particulate sensors. • Natural Disaster Web App – an interactive interface visualizing global natural-disaster data.
TECHNICAL SKILLS	Programming: Python, Julia, JavaScript, C, SQL. Modeling & computation: kinetic modeling and reaction-network simulation (Julia SciML); rate-constant inference from time-resolved NMR; full-stack scientific web applications and data pipelines.
TEACHING EXPERIENCE	<i>Massachusetts Institute of Technology</i> • Teaching Facilitator, 5.12 Organic Chemistry I Fall 2023, Spring 2024 - Average student rating: 6.5/7 <i>Carleton College</i> • Laboratory TA, Inorganic Chemistry Spring 2023 • Laboratory TA, Organic Chemistry I & II Fall 2020 – Fall 2022 • Tutor and Grader, Intermediate Chinese Fall 2020 – Fall 2022
LEADERSHIP & OUTREACH	<i>Massachusetts Institute of Technology</i> • EHS Representative and Safety Officer, Wendlandt Lab • Volunteer, MIT Women+ in Chemistry (WIC+) • Certified Mentor, Wendlandt Lab and MIT Department of Chemistry
SELECTED AWARDS	<i>Massachusetts Institute of Technology</i> • American Chemical Society, Division of Organic Chemistry, Travel Award • Department of Chemistry Mentorship Spotlight Award <i>Carleton College</i> • Jerry Mohrig Prize in Chemistry • Dean’s List • Travel Writing Contest, 3rd Place <i>Prior to Carleton College</i> • U.S. National Chemistry Olympiad, Gold Award (China Division) • Future Business Leader Summit (MicroBiz International), 3rd Place — Team CEO, Business Simulation